UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 003 · Particular integral Type 1 ($X = e^{ax}$) · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·8	Unit 2 deck 003 ("Particular integral of Type 1"), slide 8 (rendered .pptx)
T003 00:10:15	Unit 2 AV-summary transcript 003 ("Particular integral of standard functions – Type 1", 21 min), segment start
T004a 00:02:06	Unit 2 AV-summary transcript in bucket 004 ("Particular integral Type 1 and Type 2", 12.6 min); only its Type 1 part (00:00:00–00:06:04) and the constant term of its $\sin^2 x$ example are cited here. The class 004 pack cites the same file.
CW U2 P10	Unit 2 class-work problem 10
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

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1 The inverse differential operator and the particular integral

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Definitions

Non-homogeneous LDE. $f(D)y = X$ where the right-hand side X is a non-zero function of the independent variable x. S003·6 · T003 00:03:22

Inverse differential operator. $\frac{1}{f(D)}(X)$ is that function of x which is free from arbitrary constants which when operated upon by $f(D)$ gives X. S003·4 · T003 00:01:38

Particular integral (PI). A particular solution of $f(D)y = X$ containing no arbitrary constant; it is given by $\frac{1}{f(D)}(X)$. Notation PI or y_p. S003·6 · S003·13 · T003 00:05:49

General (complete) solution. $y = \text{CF} + \text{PI}$, written $y = y_c + y_p$: the complementary function (general solution of $f(D)y = 0$, with n arbitrary constants) plus the particular integral. S003·6 · S003·12 · T003 00:00:43 · T003 00:05:32

Statements in the class material

> *"The inverse differential operator is denoted as $\frac{1}{f(D)}$. $\frac{1}{f(D)}(X)$ is that function of x which is free from arbitrary constants which when operated upon by $f(D)$ gives X. Thus, $f(D)$ and $\frac{1}{f(D)}$ are inverse operators. $f(D)\left\{\frac{1}{f(D)}X\right\} = X$."* S003·4

> *"Theorem: $\frac{1}{f(D)}(X)$ is the particular solution of $f(D)y = X$."* S003·5

> *"Consider the differential equation $f(D)y = X$. The general solution of the differential equation is y = Complementary function + Particular integral. The Particular Integral is given by $\frac{1}{f(D)}(X)$."* S003·6

Key points

1. Every non-homogeneous question needs two steps: the CF from the auxiliary equation, then the PI from $\frac{1}{f(D)}X$. Write $y = y_c + y_p$ at the end. ESA ★★★ S003·6 · T003 00:17:57 · T003 00:20:36
2. The PI contains no arbitrary constant; all n constants live in the CF. That is why $\frac{1}{D}$ is integration "without writing the constant". ESA ★★ S003·4 · T003 00:04:52 · T003 00:06:24
3. Proof of the theorem: put $y = \frac{1}{f(D)}X$ in $f(D)y = X$; by definition $f(D)\left[\frac{1}{f(D)}X\right] = X$, so it is a solution, and it has no arbitrary constants, so it is a particular solution. ESA ★★ S003·5 · T003 00:03:52 · T003 00:04:17
4. The lecturer calls the inverse operator "a very important concept" and the theorem "a very important statement". ESA ★★ T003 00:01:21 · T003 00:03:15

Why?

Why can the PI have no arbitrary constants?
The constants are already all in the CF (n of them for order n). Adding more would not give a more general solution, so $\frac{1}{f(D)}X$ is defined free of constants.
S003·4 · T003 00:04:52

2 Important results: $\frac{1}{D}$ and $\frac{1}{D-a}$

Statements in the class material

> *"Some important results:* $\frac{1}{D}(X) = \int X\,dx$ *where* X *is a function of* x*.* $\frac{1}{D-a}(X) = e^{ax}\int e^{-ax}X\,dx$ *where* X *is a function of* x*. If* $f(D) = (D-a)(D-b)$ *then* $\frac{1}{f(D)}(X) = \frac{1}{(D-a)(D-b)}(X) = \frac{1}{D-a}\left(\frac{1}{D-b}(X)\right)$*."* S003·7

Key points

1. The operator always precedes the function it acts on: write $\frac{1}{f(D)}X$, never $X\cdot\frac{1}{f(D)}$. ESA ★★ T003 00:09:20
2. For $(D-a)(D-b)$ apply $\frac{1}{D-b}$ first, then $\frac{1}{D-a}$ to the result; or split into partial fractions and use E2.2 on each. ESA ★ S003·7 · T003 00:08:25 · T003 00:09:02
3. E2.2 explains the failure case: $\frac{1}{D-a}e^{ax} = e^{ax}\int e^{-ax}e^{ax}dx = xe^{ax}$, the factor x of E3.2. ESA ★★ ADDED FOR CLARITY S003·7

Exam traps

Exam trap. $\frac{1}{D^2}X$ means integrate twice. The lecture's "2 times integration" (T003 00:07:03) is not $2\int X\,dx$. S003·7 · T003 00:07:03

3 Type 1: $X = e^{ax}$ and the cases of failure

Definitions

Case of failure. $f(a) = 0$: the rule $\frac{e^{ax}}{f(a)}$ divides by zero. It happens exactly when $D - a$ is a factor of $f(D)$, i.e. a is a root of the auxiliary equation. S003·8 · T003 00:11:22

Statements in the class material

> *"Rules to find particular integral. Type (i): $X = e^{ax}$. $\frac{1}{f(D)}(e^{ax}) = \frac{e^{ax}}{f(a)}$ provided $f(a) \neq 0$. Cases of failure: when $f(a) = 0$, then $\frac{1}{f(D)}(e^{ax}) = x \cdot \frac{e^{ax}}{f'(a)}$ provided $f'(a) \neq 0$. When $f'(a) = 0$, then $\frac{1}{f(D)}(e^{ax}) = x^2 \cdot \frac{e^{ax}}{f''(a)}$ provided $f''(a) \neq 0$."* S003·8

> *"If $f(a) = 0$, then P.I $= x\frac{e^{ax}}{f'(a)}$, provided $f'(a) \neq 0$. If $f'(a) = 0$, then P.I $= x^2\frac{e^{ax}}{f''(a)}$, provided $f''(a) \neq 0$, similarly, it proceeds on when $f''(a) = 0$....."* S003·13

Key points

1. To apply Type 1, read a from the exponent: e^{-2x} has $a = -2$, a constant has $a = 0$, e^{x} has $a = 1$. Then replace every D in $f(D)$ by a. ESA ★★★ S003·8 · S003·15 · T004a 00:00:41 · T004a 00:02:59
2. $f(a) = 0$ exactly when $D - a$ is a factor of $f(D)$, i.e. when e^{ax} is already in the CF. Compare the exponent with the AE roots before substituting. ESA ★★★ T003 00:11:22 · T003 00:12:39
3. At each failure, multiply by one more x (outside the operator) and differentiate $f(D)$ once more with respect to D; stop at the first non-zero derivative. ESA ★★★ S003·8 · S003·13 · T003 00:12:08 · T004a 00:03:44
4. The power of x equals the multiplicity of a as a root: simple root x, double root x^2, triple root x^3 (Figure G3.1). ESA ★★★ ADDED FOR CLARITY T003 00:12:39 · T003 00:13:19
5. ESA: the double failure $\frac{8e^{2x}}{(D-2)^2} = 4x^2e^{2x}$ appears in three papers (Dec-23, May-24, Jun-24); the single failure $\frac{e^{2x}}{(D-1)(D-2)(D-3)} = -xe^{2x}$ in Jan-24 set 2. ESA ★★★ ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a · ESA Jan-24 s2 2d

Why?

Why does replacing D by a work?
$D^k e^{ax} = a^k e^{ax}$, so $f(D)e^{ax} = f(a)e^{ax}$. Applying $f(D)$ to $\frac{e^{ax}}{f(a)}$ therefore gives e^{ax}, which is exactly the definition of $\frac{1}{f(D)}e^{ax}$.
ADDED FOR CLARITY S003·4 · S003·8

Why an extra x when $f(a) = 0$?
Then e^{ax} solves $f(D)y = 0$ (it is in the CF), so no multiple of e^{ax} can give a non-zero right-hand side. xe^{ax} is the next independent function: $\frac{1}{D-a}e^{ax} = e^{ax}\int 1\,dx = xe^{ax}$ (E2.2).
ADDED FOR CLARITY S003·7 · T003 00:11:22

Exam traps

Exam trap. Check $f(a)$ before writing $\frac{e^{ax}}{f(a)}$: $\frac{e^{2x}}{(D-2)^2}$ is not $\frac{e^{2x}}{0}$. Compare a with the AE roots you just found for the CF. ADDED FOR CLARITY S003·8 · ESA Dec-23 3a

Exam trap. $f'(a)$ means differentiate $f(D)$ with respect to D, then put $D = a$; do not differentiate e^{ax}. The x goes outside: $x\frac{e^{ax}}{f'(a)}$. T003 00:11:51 · T003 00:12:08

Exam trap. For e^{-2x}, $a = -2$ (keep the sign): $f(-2)$ for $D^2 + 4D + 4$ is $4 - 8 + 4 = 0$, a failure. S003·15 · T004a 00:03:25

Exam trap. A constant factor stays: $\frac{8e^{2x}}{(D-2)^2} = 8 \cdot \frac{x^2e^{2x}}{2} = 4x^2e^{2x}$, not $\frac{x^2e^{2x}}{2}$. ADDED FOR CLARITY ESA Dec-23 3a

4 Extensions: constants, $\sinh ax$, $\cosh ax$ and a^x

Statements in the class material

> *"Extension of Type (i). When $X = K$:* $\frac{K}{f(D)} = K\frac{e^{0x}}{f(D)} = \frac{K}{f(0)}$. *When $X = \sinh ax$:* $\frac{\sinh ax}{f(D)} = \frac{1}{2}\left[\frac{e^{ax}}{f(a)} - \frac{e^{-ax}}{f(-a)}\right]$. *When $X = \cosh ax$:* $\frac{\cosh ax}{f(D)} = \frac{1}{2}\left[\frac{e^{ax}}{f(a)} + \frac{e^{-ax}}{f(-a)}\right]$. *When $X = a^x$:* $\frac{a^x}{f(D)} = \frac{e^{(\log a)x}}{f(D)} = \frac{e^{(\log a)x}}{f(\log a)}$. *Thus,* $\frac{a^x}{f(D)} = \frac{a^x}{f(\log a)}$."* S003·9 · S003·10

Key points

1. Rewrite the right-hand side as a sum of exponentials first, then take each term separately: $\mathrm{PI} = \mathrm{PI}_1 + \mathrm{PI}_2 + \ldots$ ESA ★★★ S003·12 · S003·16 · T003 00:19:31 · T004a 00:05:00
2. A constant is ke^{0x}: replace D by 0. If 0 is a root ($f(D)$ has no constant term), use the failure ladder with $a = 0$: $\frac{1/2}{D^3-2D^2+D} = \frac{x}{2}$. ESA ★★ S003·9 · T003 00:14:25 · T004a 00:09:42 · T004a 00:10:14 · ESA Jan-24 s3 2c
3. For a^x replace D by $\log a$, for a^{-x} by $-\log a$. Leave $\log 2$ symbolic: $\frac{2^{-x}}{(\log 2)^2+4}$. ESA ★★ S003·10 · T003 00:16:34 · T003 00:16:48 · ESA Jan-24 s1 2d
4. $\sin^2\frac{x}{2} = \frac{1-\cos x}{2}$ and $\sin^2 x = \frac{1-\cos 2x}{2}$ hide a constant $\frac{1}{2}$: a Type 1 term inside a Type 2 question. ESA ★★ ESA Jan-24 s4 2c · ESA Jan-24 s3 2c · T004a 00:09:26
5. $\sinh$ and $\cosh$ terms may fail on one exponential only: in $\frac{2\sinh x}{D^3-3D+2}$ the e^x part fails twice, the e^{-x} part not at all. ESA ★★ S003·16 · S003·17 · T004a 00:05:28 · CW U2 P11

Why?

Why is the exponent of a^x equal to $\log a$?
$a^x = e^{\log(a^x)} = e^{x\log a}$, an exponential e^{bx} with $b = \log a$. Type 1 then replaces D by $b = \log a$. S003·10 · T003 00:16:11

Exam traps

Exam trap. a^x: replace D by $\log a$, not by a. $\frac{2^{-x}}{D^2+4} = \frac{2^{-x}}{(\log 2)^2+4}$, not $\frac{2^{-x}}{8}$. (added for clarity) S003·10 · ESA Jan-24 s1 2d

Exam trap. CW P11's printed answer $\frac{x^2e^x}{5}$ is wrong: $f''(D) = 6D$, $f''(1) = 6$, so $\frac{x^2e^x}{6}$ (slide 003·17 and T004a agree). CW U2 P11 · S003·17 · T004a 00:05:28

Exam trap. Slide 003·17 calls $\mathrm{PI}_2 = \frac{e^{-x}}{4}$ but then subtracts it; keep the sign inside: $\mathrm{PI}_2 = \frac{-e^{-x}}{f(-1)} = -\frac{e^{-x}}{4}$. S003·16 · S003·17

Exam trap. Expand before splitting: $(e^{-x}+2)^2 = e^{-2x} + 4e^{-x} + 4$, and $e^{-x}\cosh x = \frac{1+e^{-2x}}{2}$ adds to both the constant and the e^{-2x} term (CW P12). (added for clarity) CW U2 P12

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). No question is Type 1 alone, but a Type 1 term sits inside 6 distinct questions (8 occurrences): $8e^{2x}$ with a double failure in Dec-23 3a (= May-24 3b = Jun-24 2a), e^{4x} in Jul-23 3b, 2^{-x} in Jan-24 set 1 2d and set 4 2c, e^{2x} with a single failure in set 2 2d, and the constant $\frac{1}{2}$ of $\sin^2$ in set 3 2c (a failure at $a = 0$) and set 4 2c. Stars: the rule and its first two failure cases ★★★; a^x, constants, $\sinh$ / $\cosh$ and $\frac{1}{D-a}$ ★★ (asked once or twice, or class-work); the third failure (x^3) ★ (lecture only, never asked).

3 · Type 1: $X = e^{ax}$ and the cases of failure

ESA Dec 2023 (7 marks) UE22MA141A Q3.a (= May 2024 Q3.b, June 2024 Q2.a) · solved in W3.5

Solve $(D-2)^2y = 8(e^{2x} + \sin 2x)$. Type 1 part: $\frac{8e^{2x}}{(D-2)^2} = 4x^2e^{2x}$ (double failure). Full solution: class 004 pack.

ESA Jul 2023 (7 marks) UE22MA141A Q3.b · solved in W3.6

Solve $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$. Type 1 part: $\frac{e^{4x}}{2}$. Full solution: class 005 pack.

ESA Jan 2024 (5 marks) UE23MA141A set 2, Q2.d · solved in W3.4

Solve $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$. Type 1 part: $-xe^{2x}$ (single failure). Full solution: class 005 pack.

4 · Extensions: constants, $\sinh ax$, $\cosh ax$ and a^x

ESA Jan 2024 (5 marks) UE23MA141A set 1, Q2.d · solved in W4.5

Solve $(D^2 + 4)y = x^2 + \cos 2x + 2^{-x}$. Type 1 part: $\frac{2^{-x}}{(\log 2)^2+4}$. Full solution: class 005 pack.

ESA Jan 2024 (6 marks) UE23MA141A set 4, Q2.c (= CW P22) · solved in W4.6

Solve $[(D-1)^2(D+1)^2]y = \sin^2\frac{x}{2} + 2^{-x} + x$. Type 1 parts: $\frac{1}{2}$ and $\frac{2^{-x}}{(\log 2)^4-2(\log 2)^2+1}$. Full solution: class 005 pack.

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q2.c (= CW P18) · solved in W4.7

Solve $(D^3 - 2D^2 + D)y = x^2e^{-x} + \sin^2 x$. Type 1 part: the constant $\frac{1}{2}$ gives $\frac{x}{2}$ (failure at $a = 0$). Full solution: class 006 pack.